

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 03

By- Vishal Sir

Recap of Previous Lecture



Topic

Logical Implication

Topic

Logical Equivalences

Topic

Argument / Inference

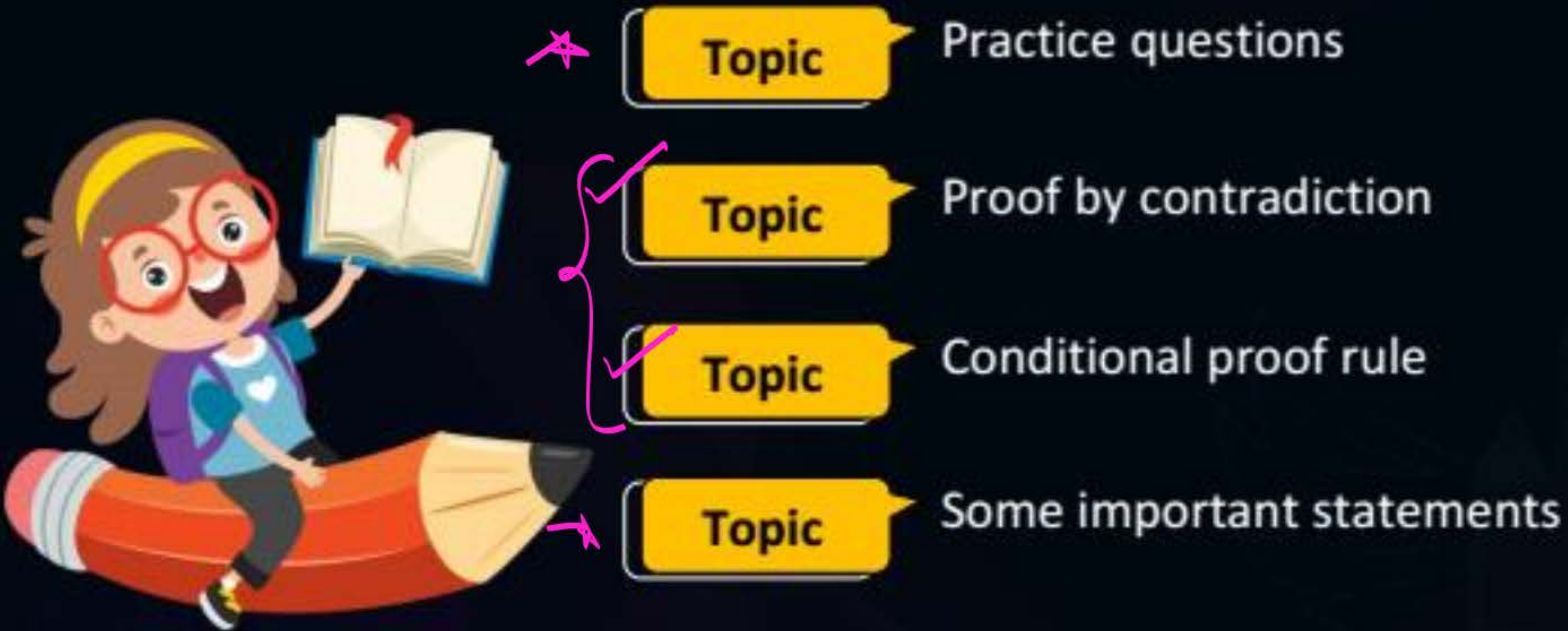
Topic

Rules of inference

Topic

Practice questions

Topics to be Covered



H.W. Q: Which of the following is/are valid?

✓ a $(P \wedge Q) \equiv T$ logically implies $(P \leftrightarrow Q) \equiv T$

✓ b $(P \leftrightarrow Q) \equiv T$ logically implies $(P \rightarrow Q) \equiv T$

✗ c $(P \leftrightarrow Q) \equiv T$ logically implies $(P \rightarrow \sim Q) \equiv T$

✓ d Q logically implies $(P \rightarrow Q)$
 $T/F \rightarrow T \equiv T$

} $\rightarrow \therefore (P \leftrightarrow Q)$ does not logically imply $P \rightarrow \sim Q$

H.W. Q: Which of the following is/are tautology?

a) $(P \wedge Q) \longrightarrow (P \longleftrightarrow Q)$ {Tautology}

b) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow Q)$ {Tautology}

~~c)~~ $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow \sim Q)$ Not a Tautology
"it is a Contingency"

d) $Q \longrightarrow (P \longrightarrow Q)$ {Tautology}

} Same as
Previous
question

H.W. Q: Which of the following is/are tautology? (valid)

a) $(a \vee b) \rightarrow (b \wedge c)$
 $T \vee F \equiv T \rightarrow \begin{matrix} T \wedge T \\ T \wedge F \\ F \wedge T \\ F \wedge F \end{matrix} \leftarrow \text{Not a tautology}$

✓ b) $(a \wedge b) \rightarrow (b \vee c)$
 $(T/F) \wedge F \equiv F \rightarrow F \vee F \equiv F \Rightarrow \therefore \text{Always true \{Tautology\}}$

c) $(a \vee b) \rightarrow (b \rightarrow c)$
 $(T/F) \vee T \equiv T \rightarrow T \rightarrow F \equiv F \therefore \text{Not a tautology}$

d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$
 $(T/F) \rightarrow T \equiv T \rightarrow T \rightarrow F \equiv F \therefore \text{Not a tautology}$

H.W. Q. Consider the following statements

$$S_1: \{ \underbrace{(A \wedge B) \rightarrow C}_P \} \equiv \{ \underbrace{(A \rightarrow C) \wedge (B \rightarrow C)}_Q \}$$

$$S_2: \{ (A \vee B) \rightarrow C \} \equiv \{ (A \rightarrow C) \vee (B \rightarrow C) \}$$

then which of the following is true ?

- (a) Both S_1 & S_2 are valid
- (b) Only S_1 is valid
- (c) Only S_2 is valid
- (d) Neither S_1 nor S_2 is valid.

$P \rightarrow Q$ must be
tautology
 $Q \rightarrow P$ must be
tautology.

H.W. Q. Consider the following statements

$$S_1: \{ \underbrace{(A \wedge B) \rightarrow C}_P \} \equiv \{ \underbrace{(A \rightarrow C) \wedge (B \rightarrow C)}_Q \}$$

$$S_2: \{ (A \vee B) \rightarrow C \} \equiv \{ (A \rightarrow C) \vee (B \rightarrow C) \}$$

$P \rightarrow Q$ must be tautology
 $Q \rightarrow P$ must be tautology.

$$\rightarrow \text{L.H.S} = (A \wedge B) \rightarrow C$$

$$\equiv \sim(A \wedge B) \vee C$$

$$\equiv (\sim A \vee \sim B) \vee C$$

$$\equiv (\sim A \vee \sim B) \vee (C \vee C)$$

$$\equiv (\sim A \vee C) \vee (\sim B \vee C)$$

$$\equiv (A \rightarrow C) \vee (B \rightarrow C)$$

$\neq$ R.H.S. $\therefore S_1$ is invalid

We can say that
 $\{ (A \rightarrow C) \wedge (B \rightarrow C) \} \rightarrow \{ (A \wedge B) \rightarrow C \}$
is a tautology

But,
 $\{ (A \wedge B) \rightarrow C \} \rightarrow \{ (A \rightarrow C) \wedge (B \rightarrow C) \}$
is not a tautology

HW Q. Consider the following statements

$$S_1: \{(A \wedge B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$$

$$S_2: \{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$$

$$\underbrace{(F \vee T) \rightarrow F}_{\text{LHS}} = F$$

$$\text{LHS} \equiv (A \vee B) \rightarrow C$$

$$\equiv \sim(A \vee B) \vee C$$

$$\equiv (\sim A \wedge \sim B) \vee C$$

$$\equiv (\sim A \vee C) \wedge (\sim B \vee C)$$

$$\equiv (A \rightarrow C) \wedge (B \rightarrow C)$$

$\neq$ R.H.S.
 $\therefore S_2$ is invalid

$P \rightarrow Q$ must be
tautology

$Q \rightarrow P$ must be
tautology.

We can observe that

$$\{(A \vee B) \rightarrow C\} \rightarrow \{(A \rightarrow C) \vee (B \rightarrow C)\}$$

is a tautology,

But

$$\{(A \rightarrow C) \vee (B \rightarrow C)\} \rightarrow \{(A \vee B) \rightarrow C\}$$

is not a tautology

H.W. Q. Consider the following statements

$$S_1: \{ \underbrace{(A \wedge B) \rightarrow C}_P \} \equiv \{ \underbrace{(A \rightarrow C) \wedge (B \rightarrow C)}_Q \}$$

$$S_2: \{ (A \vee B) \rightarrow C \} \equiv \{ (A \rightarrow C) \vee (B \rightarrow C) \}$$

then which of the following is true?

- (a) Both S_1 & S_2 are valid
- (b) Only S_1 is valid
- (c) Only S_2 is valid
- ☒ (d) Neither S_1 nor S_2 is valid.

$P \rightarrow Q$ must be
tautology
 $Q \rightarrow P$ must be
tautology.

H.W. Q. Which of the following is/are true?

~~ⓐ~~ $\{a \rightarrow (b \vee c)\} \equiv \{(a \wedge \sim b) \rightarrow c\}$

~~ⓑ~~ $\{(P \rightarrow Q) \wedge (R \rightarrow Q)\} \equiv \{(P \vee R) \rightarrow Q\}$

~~ⓒ~~ $\{(P \rightarrow Q) \vee (R \rightarrow Q)\} \equiv \{(P \wedge R) \rightarrow Q\}$

} Already proved
in previous
question

LHS: $a \rightarrow (b \vee c)$

$$\equiv \sim a \vee (b \vee c)$$

$$\equiv (\sim a \vee b) \vee c$$

$$\equiv \underbrace{\sim(a \wedge \sim b)}_{\sim x \vee c \equiv x \rightarrow c} \vee c \equiv (a \wedge \sim b) \rightarrow c \equiv \text{RHS.}$$

H.W. Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above propositional statement is.

- ☒ (a) Satisfiable but not valid { i.e., true for at least one case but not true for all }
- ☐ (b) Valid { i.e. true for all }
- ☐ (c) Contradiction { i.e. false for all }

H.W. Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

↳ R.H.S. if $R = \text{True}$

then (R.H.S.) of the given statement will become true and hence the implication will return true

True for at least one case is at least satisfiable, and not a contradiction

{ Now we need to check whether it is valid or not }

H.W. Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

If there exist at least one case in which implication returns false, then it will not be valid

→ For implication to be false, first of all
R.H.S. = $(P \wedge Q) \rightarrow R$ must be false

When $P=T, Q=T$ & $R=F$
then L.H.S. = $P \rightarrow (Q \vee R)$
i.e. $T \rightarrow (T \vee F)$
= $T \rightarrow T = T$ will be true

i.e. $P=T, Q=T$ & $R=F$ { it is the only case in which R.H.S. can be false }

So L.H.S. of main implication is true

& R.H.S. of main implication is false

∴ implication will return false

i.e. there exist at least one case for which result is false ∴ given statement is invalid

Q:- Which of the following is/are true?

(a) Conclusion R follows from the premises.

$$\{ P \rightarrow (Q \rightarrow R), (P \wedge Q) \}$$

(b) $\{ P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P \}$ logically implies S

(c) $\{ (P \vee Q), \sim P \} \vdash (P \wedge Q)$ is valid

(d) $\{ (\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S) \} \rightarrow W$ is a tautology

Q:- Which of the following is/are true?

(a) Conclusion R follows from the premises.

$\{ P \rightarrow (Q \rightarrow R), (P \wedge Q) \}$

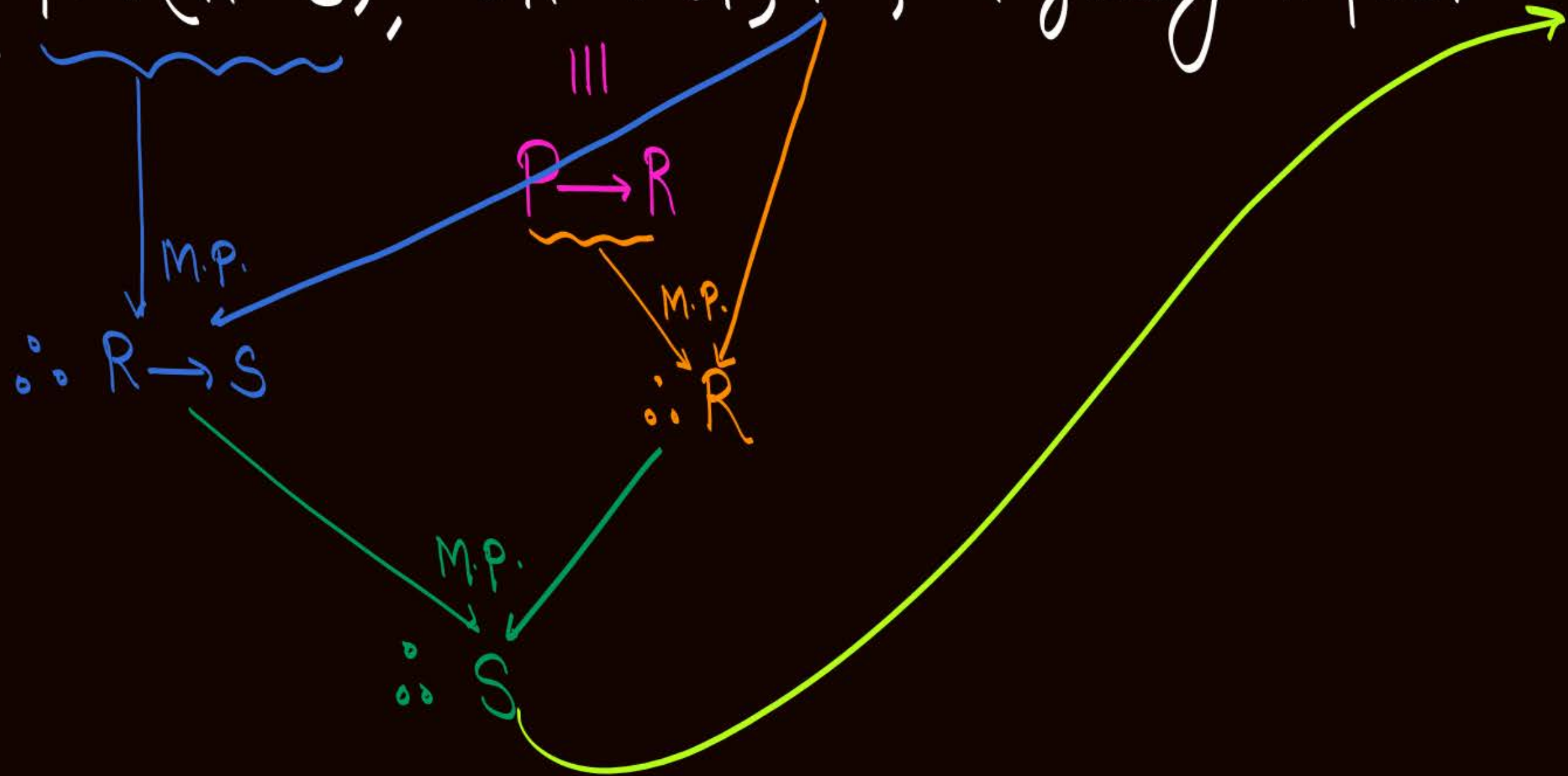
||| Simplification

$\underbrace{P, Q}$
Modus Ponens
 $\therefore Q \rightarrow R$

$\underbrace{Q \rightarrow R, Q}_{m.p.} \therefore R$ } i.e. 'R' follows from the given premises

Q:- Which of the following is/are true?

(b) $\{ P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P \}$ logically implies S



Q:- Which of the following is/are true?

© $\{ (P \vee Q), \sim P \} \vdash (P \wedge Q)$ is valid

$\downarrow$
 $\sim P$ is true
i.e., 'P' is false

$\rightarrow$ then

$P \wedge Q$ Can never be true
 $\therefore$ given statement is false

Q:- Which of the following is/are true?

(d) $\{ (\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S) \} \rightarrow W$ is a tautology

Check if W follows from $\{ (\sim T \rightarrow \sim R), \sim S, (T \rightarrow W), (R \vee S) \}$ or not

$$\begin{array}{c} \text{III} \\ R \rightarrow T \end{array}$$

m.p.

$\therefore$

$\vdash$

m.p.

$\therefore W$

Disjunctive Syllogism

$\therefore R$

$\therefore$ i.e., W follows from the set of premises

Q:- Which of the following is/are true?

✓ (a) Conclusio R follows from the premises.

$$\{ P \rightarrow (Q \rightarrow R), (P \wedge Q) \}$$

✓ (b) $\{ P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P \}$ logically implies S

~~(c) $\{ (P \vee Q), \sim P \} \vdash (P \wedge Q)$ is valid~~

✓ (d) $\{ (\sim T \rightarrow \sim R) \wedge \sim S \wedge (T \rightarrow W) \wedge (R \vee S) \} \rightarrow W$ is a tautology

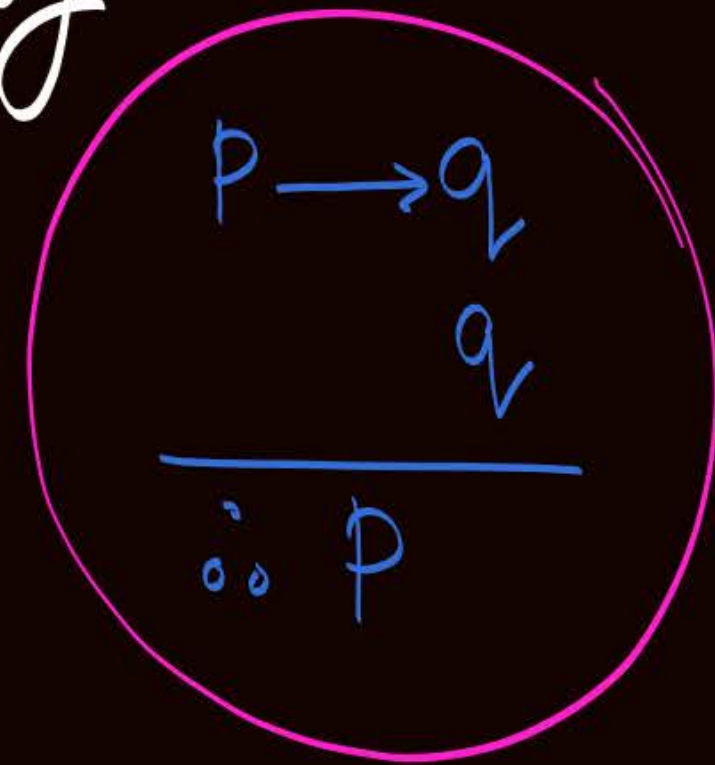
Q:- Check whether the following argument is valid or invalid

* If today is Vishal's B'day, then today is 1st April

* Today is 1st April

∴ Today is Vishal's B'day

let, P : Today is Vishal's Birthday } ∴ Argument is
 Q : Today is 1st April



it is invalid
{ Fallacy of affirming the consequent }

Q:- Check whether the following argument is valid or invalid

* If today is Vishal's B'day, then today is 1st April

* Today is not 1st April

∴ Today is not Vishal's B'day

* p : Today is Vishal's Birthday

q : Today is 1st April

∴ $\sim q$: Today is not the 1st April

∴ Argument is,

$p \rightarrow q$

$\sim q$

∴ $\sim p$

it is Valid
{ Modus Tollens }

Q: Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies $\sim P$

$(\sim P \vee Q), (\sim P \vee \sim Q)$

$$\frac{P \vee Q}{\sim P \vee \sim Q}$$

Resolution

at least one of Q or $\sim Q$ is false
i.e. by resolution

$\sim P \vee \sim P \equiv \sim P$ is true

Hence,
argument is
"valid"



Topic : Proof by contradiction

If we want to check whether the argument,

P_1
 P_2
 $\vdots$
 P_n

∴ Q
is valid or not

then, in order to check for the validity of the argument,

We will assume that the Conclusion of the argument is false. { i.e, we will Assume that Q is false, i.e, " $\sim Q$ " is true }, and we will include that false Conclusion in the set of premises.

∴ Our new set of premises will be
 $\{ P_1, P_2, \dots, P_n, \sim Q \}$

If there is no Contradiction then Argument is invalid

If this assumption results in any Contradiction, then our assumption is wrong { we assume that Conclusion is false and our assumption is wrong, i.e; our Conclusion is true }
∴ And hence our argument is a valid argument



Topic : Some important statements

1. $P \rightarrow Q$
2. P implies Q
3. If P then Q
4. P only if Q
5. P is sufficient condition for Q
6. Q is necessary condition for P



Topic : Some important statements

7. P if Q

8. P when Q

9. P follows from Q

10. P unless $\sim Q$



2 mins Summary



Topic

Practice questions

Topic

Proof by contradiction

Topic

Conditional proof rule

Topic

Some important statements

THANK - YOU